Chapter 7, Problem 31P

Problem

Evaluate the following integrals:

(a) $\int_{-\infty}^{\infty} e^{-4t^2}\delta(t-2)dt$

(b) $\int_{-\infty}^{\infty} [5\delta(t) + e^{-t}\delta(t) + \cos 2\pi t\delta(t)]dt$

Step-by-step solution

Step 1 of 4

(a)

Evaluate the integral,

$\int_{-\infty}^{\infty} e^{-4t^2}\delta(t-2)dt$ ...... (1)

The integral contains delta function. So, the integral is non-zero at certain instants of time and zero otherwise.

Consider the integral with delta function.

$\int_{-\infty}^{\infty} f(t)\delta(t-a)dt = f(a)$ ...... (2)

Step 2 of 4

Compare equation (1) and (2).

$$f(t) = e^{-4t^2}$$
$$a = 2$$

Evaluate the integral, $\int_{-\infty}^{\infty} e^{-4t^2}\delta(t-2)dt$.

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-4t^2}\delta(t-2)dt &= f(2)\\ &= \left[e^{-4t^2}\right]_{t=2}\\ &= e^{-4\times2^2}\\ &= 112\times10^{-9}\end{aligned}$$

Thus, the value of $\int_{-\infty}^{\infty} e^{-4t^2}\delta(t-2)dt$ is $\boxed{112\times10^{-9}}$.

Step 3 of 4

(b)

Evaluate the integral,

$$\int_{-\infty}^{\infty}\left[5\delta(t)+e^{-t}\delta(t)+\cos 2\pi t\delta(t)\right]dt=\int_{-\infty}^{\infty}\left[5+e^{-t}+\cos 2\pi t\right]\delta(t)\,dt \quad \ldots\ldots (3)$$

The integral contains delta function. So, the integral is non-zero at certain instants of time and zero otherwise.

Consider the integral with delta function.

$$\int_{-\infty}^{\infty} g(t)\delta(t-b)\,dt = g(b) \quad \ldots\ldots (4)$$

Step 4 of 4

Compare equation (3) and (4).

$$g(t)=5+e^{-t}+\cos 2\pi t$$
$$b=0$$

Evaluate the integral, $\int_{-\infty}^{\infty}\left[5\delta(t)+e^{-t}\delta(t)+\cos 2\pi t\delta(t)\right]dt$.

$$\begin{aligned}\int_{-\infty}^{\infty}\left[5\delta(t)+e^{-t}\delta(t)+\cos 2\pi t\delta(t)\right]dt &= g(0)\\ &=\left[5+e^{-t}+\cos 2\pi t\right]_{t=0}\\ &=5+e^{0}+\cos 0\\ &=7\end{aligned}$$

Thus, the value of $\int_{-\infty}^{\infty}\left[5\delta(t)+e^{-t}\delta(t)+\cos 2\pi t\delta(t)\right]dt$ is $\boxed{7}$.